

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

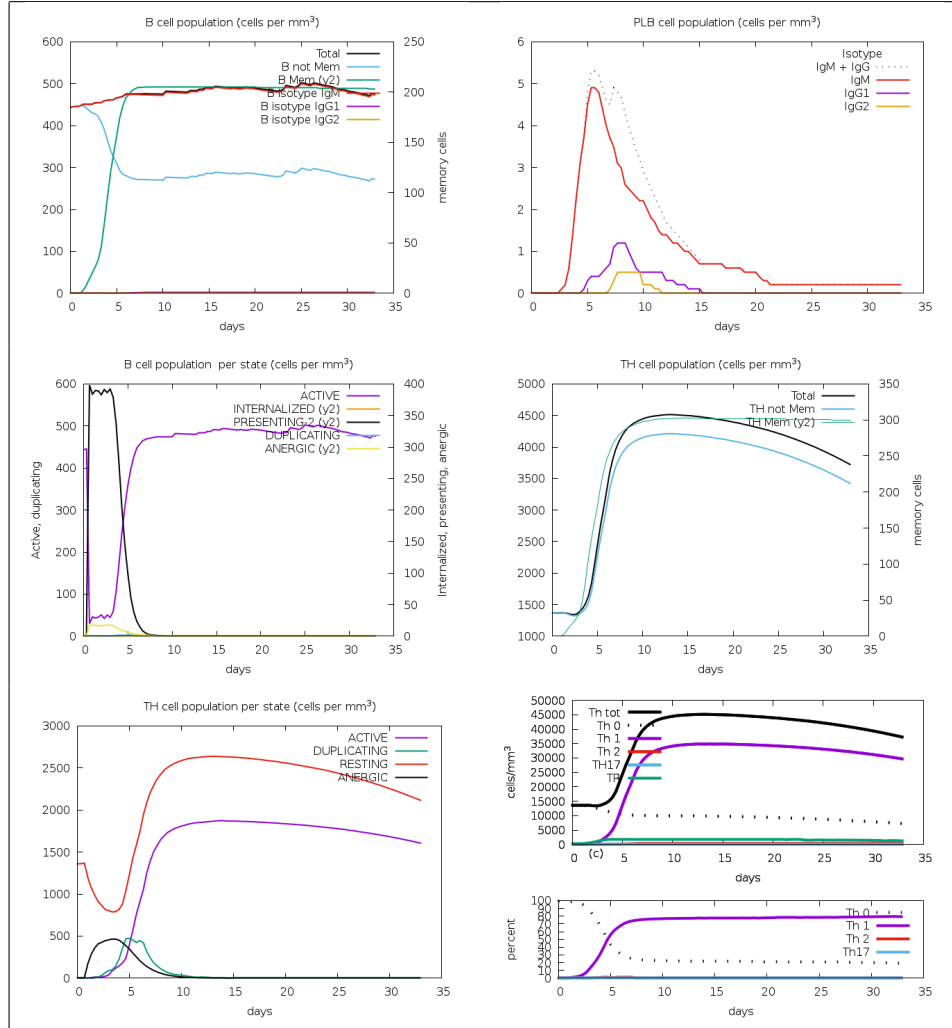


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

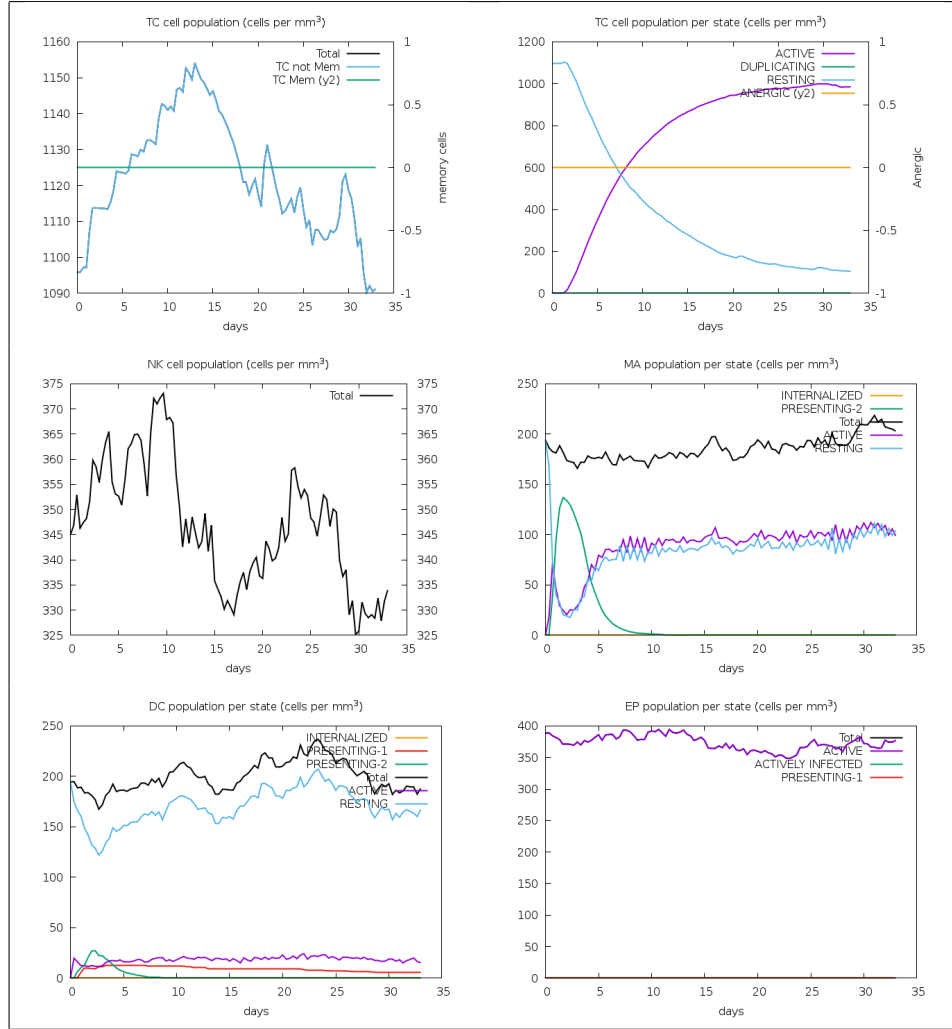


Figure 2: Legend: symbols as figure above.

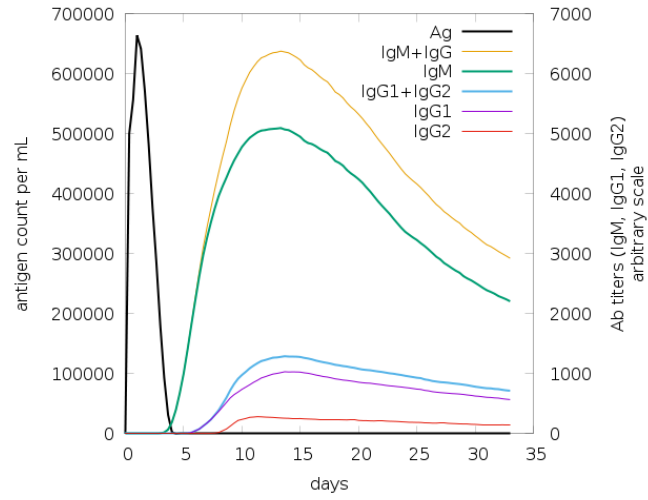


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

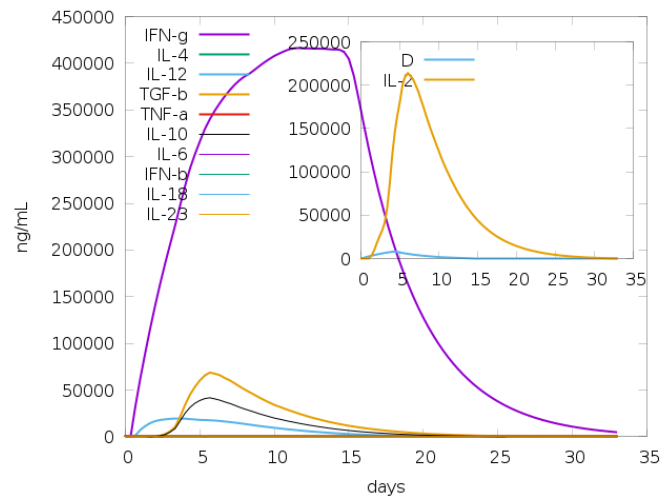


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

APPHALSGSDQATRVYQDLGGPSQAPEAAAKPGTGPNGNLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGG
RPGAGGAGGSPRHRRGVRRPKRPSCICVCKLTHGGTGACAGGAGGAGGAGGAGGAGGAGGAGGAGGAGGAGGAGGAGGAGA
GGAGGAGGAGGAGGAGGAEEAAKRVLSVGLGTGGTGMPTFEVSVAFGPGPGACMLSAPLEKPGPGPLQGQLTAYGPPGP
FLDKGYTYLLGGPGGVFSDETPASDDMPTNTEAAKKPMVEGAAAEAGDGDGPGPGSDREVRGSALYEAST

[illegible]

```

1]      pos=7 len=6      GGSDQA
2]      pos=23 len=4     SQAP
3]      pos=32 len=6     GPGTGP
4]      pos=41 len=53    LGEKGDTSGPEGSGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGSGSGPRH
5]      pos=95 len=4     DGVR
6]      pos=111 len=66   GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGAGGGAGAGGGAGGAGAGGGAG
7]      pos=264 len=4    DMPT
8]      pos=282 len=19   AAEGDDGDGPGPGSDREVR

```

[illegible][illegible]

```

1]      pos=41 len=6      GPGTGP
2]      pos=50 len=53     LGEKGDTSGPEGSGSGPQRRGGDNHHRGRGRGRGRGGGRPGAPGSGSGSPRH
3]      pos=104 len=4     DGVR
4]      pos=120 len=66     GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGAGGGAG
5]      pos=295 len=20     AAEGDDGDGPGPGSDREVRG

```

AKFVAAWTLKAAAGGSDLERGGPPRRRPPRGPPLSGP GTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGR
GGGRPGAPGSGSGSGPRHRDGVRRPQKRPSICGCKGTHGGGTACAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGG
AGAGGGAGGAGGAGAGGAGGAGAAEAALKVPKNVEVWGPGPGRQIPKIQNFPGPGMVFLQTHIFAEVLKDGPGPGSALIAG
GSGILQTNFKEAAAKNEIFLTKKMTGVCQA

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_007

[illegible]

```

1]      pos=58 len=4      AGTD
2]      pos=153 len=9     AAEGDDGDA
3]      pos=205 len=7     ASDREVR
4]      pos=270 len=6     ADGKNK
5]      pos=301 len=4     DMPT

```

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_008

[illegible]

```

1]      pos=113 len=4      AGTD
2]      pos=144 len=7      ASDREVR
3]      pos=232 len=9      AAEGDDGDA
4]      pos=255 len=4      DMPT
5]      pos=265 len=6      ADGKNK

```

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_009

[illegible]

GGGAGGAGGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQLTAYGPGPG
FLDKGTYTLGPGPGVFSDEIPASQDMPTNTEAAAKPPMVEGAAAEGDDGDGPGPGSDREVRGSALYEAST

Epitopes of protein 0 -----

0] pos= 8 score=0.052333 unnormalised=5.7896000000 GSDQATRVY
1] pos= -1 score=0.947667 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_002

AKFVAAWTLKAAAGGSCPSLDVDSIIPRTPDVLHEDEAAAKGPGTGPNGLGEGDTS GPEGSGGSGPQRRGGDNHGRGR
GRGRGRGGGRPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGG
AGAGGGAGAGGGAGGAGGAGAGGAGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQ
LTAYGPGPGFLDKGTYTLGPGPGPLEKQLFYYIGTMLPNTRPHEAAAKPPMVEGAAAEGDDGDGPGPGSDREVRGS

Epitopes of protein 1 -----

0] pos= 263 score=0.033756 unnormalised=3.6626000000 PLEKQLFYY
1] pos= -1 score=0.966244 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_003

AKFVAAWTLKAAAGGSDLERGPPGPRRPPRPGPLSGPGTGPNGLGEGDTS GPEGSGGSGPQRRGGDNHGRGRGRGRGR
GGGRPGAPGGSGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGGAGGAGGAGGAGGAGG
AGAGGGAGGAGGAGGAGGAGAEAAAKVPKNVEVWGPGRQIPKIQNFGPGPMVFLQTHIFAELVDGPGPGSALIA
GSILQTNFKEAAKNEIFLTKKMTVCQA

Epitopes of protein 2 -----

0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_004

AKFVAAWTLKAAAGSGPGTGPNGLGEGDTS GPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRPPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGAGGGAGGAGGAGGAGGAGGAGGGA
GAEEAAKFLDKGTYTLGPGPGTMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 3 -----

0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_005

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AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKMQSYAAAAAIALYLQNNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAASCFTASVSTVVSATAAAAAATPDIL
PVVTRNLNAI

Epitopes of protein 4 -----

0] pos= 14 score=0.052226 unnormalised=5.7896000000 GSDQATRVY
1] pos= 136 score=0.002044 unnormalised=0.2266000000 AAAAAIALY
2] pos= -1 score=0.945730 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_006

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_011

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGAGAGGAGAGGAGAGAGAGGAGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGPLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDPGPGSDREVRGSALYEASTGPG

Epitopes of protein 10 -----

0] pos= 247 score=0.033756 unnormalised=3.662600000 PLEKQLFYY
1] pos= -1 score=0.966244 unnormalised=104.841000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_012

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGAGAGGAGAGGAGAGAGAGGAGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAANKPPM

Epitopes of protein 11 -----

0] pos= -1 score=1.000000 unnormalised=104.841000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_013

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGAGAGGAGAGGAGAGAGAGGAGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAANKPPM

Epitopes of protein 12 -----

0] pos= -1 score=1.000000 unnormalised=104.841000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_014

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGAGAGGAGAGGAGAGGAGAGAGGAGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGNTTVGIELPDAFKCEAAAKPPMVEGAAAEAGDDGDPGPGSDREVRGSALYEASTGPGPGLEI

Epitopes of protein 13 -----

0] pos= -1 score=1.000000 unnormalised=104.841000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_015

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGAGAGGAGAGGAGAGAGAGGAGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGNKCSQGA VAGEPRQIPKIQNFTRTQKSCIEAAAKPPMVEGAAAEAGDDGDPGPGSDREVRGSA

Epitopes of protein 14 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_004

AKFVAAWTLKAAAGSGPGTGPNGLGKEDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGGA
GAEEAAKFLDKGTYTLGPGPGTMEIAGLYGPGPGTSKFLMGTYKRVTEKPGPGPRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 3 -----

0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_005

AKFVAAWTLKAAAGSDQATRVYQDLGGPSQAPPIDELAQHASGEGPGINPISKTAALALRGPPGPRRPPRGPPPLS
AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAIALYLQNNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAASCFTASVSTVVSATAAAAAATPDIL
PVTTRNLNAI

Epitopes of protein 4 -----

0] pos= 14 score=0.052226 unnormalised=5.7896000000 GSDQATRVY
1] pos= 136 score=0.002044 unnormalised=0.2266000000 AAAAAIALY
2] pos= -1 score=0.945730 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_006

AKFVAAWTLKAAAGSGPIDELAQHASGEGPGINPISKTAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTTRNLNAIAAAAAIALYLQNNWAAAAATVKGMQSYAAAAADPYQVPFVAAAAALRGPPGPRRPPRGPPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 5 -----

0] pos= 43 score=0.030800 unnormalised=3.3916000000 AADQATRVY
1] pos= 171 score=0.002058 unnormalised=0.2266000000 AAAAAIALY
2] pos= 198 score=0.015053 unnormalised=1.6576000000 YAAAAADPY
3] pos= -1 score=0.952089 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_007

AKFVAAWTLKAAAGSVTVTAFWAAAAAYTIPWTELMAAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEVPDIPAVAAAAAIFLIGFALFAAAAAVEIPEFPFYAAAAAKLPHWPTLAAAAAPPMVEGAAAEGDDG
DAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAASDREVRSALYEASTAAAAATSSFVTNTTVGIEL
PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFHERADSFAAAAAVFSDEIPASQDMPTNTAAAAAGEGNWTS
L DGGTFK

Epitopes of protein 6 -----

0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_008

AKFVAAWTLKAAAGSYTIPWTELMAAAAAVTAFWAAAAAESLWGVYRAAAAIIFLIGFALFAAAAAANVFDLEGIF
RAAAAAKLPHTPTLAAAAAVEIPEFPFYAAAAAGTDNEIFLTAAAAAALAEVPDIPAVAAAAASDREVRSALYEAST
AAAAATSSFVTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAPPMVEGAAAEGDDG
AAAAAVFSDEIPASQDMPTNTAAAAADGKNKETFHERADSFAAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
LD GGTfK

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

APPHALSGGSAPTGSWFSAPQAPENAYQAYAAAPQLFPVSDITQNNQNTNAGGEAPQPGDNSTVQEAALKGPGTGPNGNL
GEKGDTSGPEGSGGGSPQRRGCDNHGRGRGRGRGGGRPGAPGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAG
AGGAGAGGAGAGGGAGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGGAGGAGAGGGAGAAAKRVLSVVVLGPGPGTMFEV
SVAFGPGPGACMLSAPLEKGPGLPQQGLTAYGPGPGFLDKGTYTLGPGPGSLEREDRAWHLPAAAA

```
0]      pos=  -1 score=1.000000 unnormalised=104.8410000000      non-binding event
```

APPHALSGGSAPTGSWFSAPQAPENAYQAYAAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAALKPGTGPNGNL
GEKGDTSGPEGSGGGSPQRRGGDNHGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAG
AGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGGAGGAGAGGGAGAGGGAGAGGGAGAEAAAKRVLSVVVLGPGPGTMFEV
SVAFGPGPGACMLSAPEKGPGLPQGGQLTAYGPGPGFLDKGTYTLGPGPGLTPHTKAVYQPRGAFGGS

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

APPHALSGGSPEPLPQQGLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGRRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLVVLLGPGPCTMFEVSVAFGPGPGACMLSAPELEKGPGLPQQGLTAYGPGPGFLDKGTY
TLGPGPGLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEASTGPG

```
0] pos= 247 score=0.033756 unnormalised=3.6626000000 PLEKQLFYY
1] pos= -1 score=0.966244 unnormalised=104.8410000000 non-binding event
```

APPHALSGGSPEPLPQQGLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGRRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAAAKRVLVVLLGPGPTMFEVSVAFGPGPGACMLSAPELEKGPGLPQQGLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNQNTNQAGGEAPQPGDNSTVQEAALKPPM

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

APPHALSGGSPEPLPQQGLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGRRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPELEKGPGLPQQGLTAYGPGPGFLDKGTY

Epitopes of protein 12 -----

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGNLGEKGDTSGPEGSSGGSPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGRPHRHDGVRPPKKRPSCIGCKCTHTGGTGACAGAGGACAGGAGGAGGAGGAGGAGGAGGAGGAGGAGGAGGAGGAGGA
GGAGAGCGGAGAGAEEAARLVSVVLLLPMPGVETAEFVSVAFGPGPACMLSAPELKGGPGLPQQLTYAGPGPGPGLDKGT
YTLGPGGNNTVGIELPDFAFKCAAALKPGPVTEMAEAGDDGDPGPGSDREVRKSALYEASTGPPGPELEI

Epitopes of protein 13 -----

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPESGGSGPQRRGGDNHGRGRGRGRGGGRRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPELKGPGPLPQQQLTAYGPGPGFLDKGTY
TLGPGPGNKCSQGAVAGEPRQIPKIQNETRTQKSCIEAAAKPPMVEGAAAEAGDDGDGPGPSDREVRGSA

Epitopes of protein 14 -----

[illegible]

Epitopes of protein 15 -----

=====

Max score: 28.406000

APPHALSGGSDQATRVYQDLGGPSQAPEAAAKPGTGPNGLGEKGDTSGPEGSGGSGPQRRCNDHNRGRGRGRGRGGG
RPGAPGSGSGSPRRHRDGVRRPQKRPSICGCKGTHTGGTGAGAGAGGAGAGGAGGAGGAGGAGGAGGAGGAGGAGGAGA
GGGAGGAGGAGAGGGAGAEAAAKRVLVSVVLLGPGPTMFEVSVAFGPGPGACMLSAPLEKGP GPGLPQQQLTAYGPGPG
FLDKGYTLGPGPVFSDEIPASQDMPTNTEAAKKPPMVEGAEEAGDDGDGPGPGSDREVRSALYEAST

Epitopes of protein 0 -----

1]	pos=	51	score=0.036028	unnormalised=4.5212000000	NPISKTA
2]	pos=	72	score=0.019844	unnormalised=2.4902000000	PPRGPPLSA
3]	pos=	75	score=0.009341	unnormalised=1.1722000000	GPPLSAAAA
4]	pos=	103	score=0.025924	unnormalised=3.2532000000	VPFVAAAAA
5]	pos=	117	score=0.032402	unnormalised=4.0662000000	APQLFAAAA
6]	pos=	135	score=0.008592	unnormalised=1.0782000000	YAAAAAIAL
7]	pos=	205	score=0.019923	unnormalised=2.5002000000	IPNLFAAAA
8]	pos=	-1	score=0.835449	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_006

AKFVAAWTLKAAAGGSPGIDELAQHASGEGPGINPISKTAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKLSSTEFIPNLFAAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNIAAAAAAIALVLQGNWAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 5 -----

0]	pos=	34	score=0.035294	unnormalised=4.5212000000	NPISKTA
1]	pos=	56	score=0.038627	unnormalised=4.9482000000	GPSQAPAAA
2]	pos=	72	score=0.031742	unnormalised=4.0662000000	APQLFAAAA
3]	pos=	126	score=0.019517	unnormalised=2.5002000000	IPNLFAAAA
4]	pos=	170	score=0.002421	unnormalised=0.3102000000	IAAAAAIAL
5]	pos=	208	score=0.025395	unnormalised=3.2532000000	VPFVAAAAA
6]	pos=	227	score=0.019439	unnormalised=2.4902000000	PPRGPPLSA
7]	pos=	230	score=0.009150	unnormalised=1.1722000000	GPPLSAAAA
8]	pos=	-1	score=0.818414	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_007

AKFVAAWTLKAAAGGSVTVTAFWAAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEVDIPAVAAAAAIFLIGFALFAAAAAVEIPEFFFYAAAAAKLPHWPTLAAAAAPPMVEGAAAEGDDG
DAAAAADPPFNSILFALLAAAAAANFKLSSTEFIPNLFAAAAAADREVRSALYEASTAAAAATSSFTNTTVGIEL
PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFHERADSFAAAAAVFSDEIPASQDMPTNTAAAAAGEGNWTS
DGGTFK

Epitopes of protein 6 -----

0]	pos=	95	score=0.037638	unnormalised=5.0192000000	IPAVAAAAA
1]	pos=	120	score=0.006151	unnormalised=0.8202000000	IPEFFFYAA
2]	pos=	123	score=0.036836	unnormalised=4.9122000000	FPFYAAAAA
3]	pos=	133	score=0.008183	unnormalised=1.0912000000	LPHWPTTLA
4]	pos=	137	score=0.026660	unnormalised=3.5552000000	TPTLAAAAA
5]	pos=	140	score=0.009983	unnormalised=1.3312000000	LAAAAAPPM
6]	pos=	145	score=0.000774	unnormalised=0.1032000000	APPMVEGAA
7]	pos=	196	score=0.018749	unnormalised=2.5002000000	IPNLFAAAA
8]	pos=	221	score=0.013529	unnormalised=1.8042000000	AAAAATSSF
9]	pos=	296	score=0.015269	unnormalised=2.0362000000	IPASQDMPT
10]	pos=	302	score=0.040038	unnormalised=5.3392000000	MPTNTAAAA
11]	pos=	-1	score=0.786190	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_008

AKFVAAWTLKAAAGGSYITIPWTELMAAAAAVTTFWAAAAAASLWVGVYRAAAAIFLIGFALFAAAAAANVFDLEGIF
RAAAAAAKLPHWPTLAAAAAAVEIPEFFFYAAAAAGTDNEIFLTAAAAAALAEVDIPAVAAAAADREVRSALYEAST
AAAAATSSFTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKLSSTEFIPNLFAAAAAAPPMVEGAAAEGDDGD
AAAAAVFSDEIPASQDMPTNTAAAAADGKNKETFHERADSFAAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
DGGTFK

0]	pos= 87	score=0.008238	unnormalised=1.0912000000	LPHWPTTLA
1]	pos= 91	score=0.026840	unnormalised=3.5552000000	TPTLAAAAA
2]	pos= 102	score=0.006192	unnormalised=0.8202000000	IPEFFFYAA
3]	pos= 105	score=0.037084	unnormalised=4.9122000000	FPFYAAAAA
4]	pos= 136	score=0.037892	unnormalised=5.0192000000	IPAVAAAAA
5]	pos= 160	score=0.013621	unnormalised=1.8042000000	AAAAATSSF
6]	pos= 215	score=0.018875	unnormalised=2.5002000000	IPNFLAAAA
7]	pos= 219	score=0.003316	unnormalised=0.4392000000	FAAAAAAPM
8]	pos= 224	score=0.000779	unnormalised=0.1032000000	APPMVEGAA
9]	pos= 250	score=0.015372	unnormalised=2.0362000000	IPASQDMPT
10]	pos= 256	score=0.040308	unnormalised=5.3392000000	MPTNTAAAA
11]	pos= -1	score=0.791485	unnormalised=104.8410000000	non-binding event

APPHALSGGSAPTGSWFSAPQPAENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKGPGTGPNGL
GEKGDTSPEGSGSGSPQRRCGDNHGRGRGRGRGRGGRRPGAPGSGSGSPRHRDVRRPQKRPSICGCKGTHGGTGAGAG
AGGAGAGGAGAGGAGAGGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEEAAAKRVLSVVLLGPGPGTMFEV
SVAFGPGPGACMLSAPLEKGPGPGLPOGQLTAYCGPGPFGLDKGTYTLGPGPGSLEREDRAWHLPAAAA

0]	pos= 10	score=0.021143	unnormalised=2.3742000000	APTGSWFSA
1]	pos= 18	score=0.002664	unnormalised=0.2992000000	APQPAPENA
2]	pos= 54	score=0.008711	unnormalised=0.9782000000	APQPGDNST
3]	pos= 118	score=0.002682	unnormalised=0.3012000000	RPGAPGSSG
4]	pos= 137	score=0.017207	unnormalised=1.9322000000	RPQKRPSCI
5]	pos= 261	score=0.013956	unnormalised=1.5672000000	GPGLPQQGL
6]	pos= -1	score=0.933636	unnormalised=104.8410000000	non-binding event

APPHALSGGSAPTGSWFAAQPAENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKGPGTGPNGNL
GEKGDTSGPEGSGGSGPQRGRGDNHGRGRGRGRGGRRPGAPGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAG
AGGAGAGGAGAGGAGAGGAGGAGGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAEEAAAKRVLSSVVLLGPGPGMTFEV
SVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGYTLGPGPGLTPHTHKAVYQPRGAFFGGS

0]	pos=	10	score=0.021049	unnormalised=2.3742000000	APTGSWFSA
1]	pos=	18	score=0.002653	unnormalised=0.2992000000	APQPAPENA
2]	pos=	54	score=0.008672	unnormalised=0.9782000000	APQPGDNST
3]	pos=	118	score=0.002670	unnormalised=0.3012000000	RPGAPGGSG
4]	pos=	137	score=0.017130	unnormalised=1.9322000000	RPQKRPSCI
5]	pos=	261	score=0.013894	unnormalised=1.5672000000	GPGLPQGQL
6]	pos=	298	score=0.004452	unnormalised=0.5022000000	AVYQPRGAF
7]	pos=	-1	score=0.929479	unnormalised=104.8410000000	non-binding event

APPHALSGGSPPELPQQQLTEAAAKGPGTGPNGLGEKGTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRRPGAPGG
SGSGPRHRDGVRRPQKRPSCIGCKTHTGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLVVVLLGPGPGTMFEVSVAFGPGPGACMLSAPEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGPLEKQLEYIYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALEYASTGPG

20

0]	pos=	13	score=0.005907	unnormalised=0.6622000000	LPQGQLTEA
1]	pos=	73	score=0.002687	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	92	score=0.017237	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.013980	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	260	score=0.024935	unnormalised=2.7952000000	LPNTRPHEA
5]	pos=	-1	score=0.935254	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_012

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 11 -----

0]	pos=	13	score=0.005862	unnormalised=0.6622000000	LPQGQLTEA
1]	pos=	73	score=0.002667	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	92	score=0.017106	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.013875	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	247	score=0.021019	unnormalised=2.3742000000	APTGSWFSA
5]	pos=	255	score=0.002649	unnormalised=0.2992000000	APQPAPENA
6]	pos=	291	score=0.008660	unnormalised=0.9782000000	APQPGDNST
7]	pos=	-1	score=0.928163	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_013

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 12 -----

0]	pos=	13	score=0.005862	unnormalised=0.6622000000	LPQGQLTEA
1]	pos=	73	score=0.002667	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	92	score=0.017106	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.013875	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	247	score=0.021019	unnormalised=2.3742000000	APTGSWFSA
5]	pos=	255	score=0.002649	unnormalised=0.2992000000	APQPAPENA
6]	pos=	291	score=0.008660	unnormalised=0.9782000000	APQPGDNST
7]	pos=	-1	score=0.928163	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_014

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGNTTVGIELPDAFKCEAAAKPPMVEGAAAEAGDDGDPGPGSDREVRGSALEYASTGPGPGELEI

Epitopes of protein 13 -----

0]	pos=	13	score=0.006040	unnormalised=0.6622000000	LPQGQLTEA
1]	pos=	73	score=0.002747	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	92	score=0.017623	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.014294	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	242	score=0.003057	unnormalised=0.3352000000	GPGPGNTTV
5]	pos=	-1	score=0.956238	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_015

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGRRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGGAGAGGAGGAGAGAGAGGGAGAGGGAGGA
GGAGAGGGAGAAEAARVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGT
TLGPGPGNKCQGAAGEPRQIPKIQNFRTRTQKSCIEAAAKPPMVEGAAEGDDGDGPGPGSDREVRGSA

Epitopes of protein 14 -----

0]	pos=	13	score=0.006058	unnormalised=0.6622000000	LPQQQLTEA
1]	pos=	73	score=0.002756	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	92	score=0.017677	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.014338	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	-1	score=0.959171	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_016

AKFVAAWTLKAAAGGSANQQQLADIGVPQAPVAAPARRTRKPQQPESLEECDSELEIKRYKNRVASEAAAKGPGTGPG
NGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGRRPGAPGGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGAGAGGGAGGAGAGGGAGAGGAGAAEAARVLSVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTQTQPEAAAKPPMVE

Epitopes of protein 15 -----

0]	pos=	27	score=0.033098	unnormalised=3.8422000000	VPQPAPVAA
1]	pos=	29	score=0.031022	unnormalised=3.6012000000	QPAPVAAPA
2]	pos=	121	score=0.002595	unnormalised=0.3012000000	RPGAPGGSG
3]	pos=	140	score=0.016645	unnormalised=1.9322000000	RPQKRPSCI
4]	pos=	264	score=0.013500	unnormalised=1.5672000000	GPGLPQQQL
5]	pos=	-1	score=0.903140	unnormalised=104.8410000000	non-binding event

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Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE

Threshold: 8.702800

Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_001

APPHALSGGSDQATRVYQDLGGPSQAPEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGG
RPGAPGGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGAGGAGAGGAGAGGGAGA
GGGAGGAGGAGAGGGAGAAEAARVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPG
FLDKGTYTLGPGPGVFSDEIPASQDMPTNTTEAAAKPPMVEGAAEGDDGDGPGPGSDREVRGSALYEAST

Epitopes of protein 0 -----

0]	pos=	21	score=0.020802	unnormalised=2.4282000000	GPSQAPEAA
1]	pos=	80	score=0.002580	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	99	score=0.016553	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	223	score=0.013426	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	259	score=0.017444	unnormalised=2.0362000000	IPASQDMPT
5]	pos=	265	score=0.031023	unnormalised=3.6212000000	MPTNTEAAA
6]	pos=	-1	score=0.898171	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_002

Epitopes of protein 1 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_003

Epitopes of protein 2 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_004

Epitopes of protein 3 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_005

Epitopes of protein 4 -----

23

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_006

AKFVAAWTLKAAAGGSPGIDELAQGHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKSLSSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNIAAAAAIALYLQKNWAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPPLSAAAA
ASCFTASVSTVVSAT

Epitopes of protein 5 -----

0]	pos=	34	score=0.035294	unnormalised=4.5212000000	NPISKTA
1]	pos=	56	score=0.038627	unnormalised=4.9482000000	GPSQAPAAA
2]	pos=	72	score=0.031742	unnormalised=4.0662000000	APQLFAAAA
3]	pos=	126	score=0.019517	unnormalised=2.5002000000	IPNLFAAAA
4]	pos=	170	score=0.002421	unnormalised=0.3102000000	IAAAAAIAL
5]	pos=	208	score=0.025395	unnormalised=3.2532000000	VPFVAAAAA
6]	pos=	227	score=0.019439	unnormalised=2.4902000000	PPRGPLSA
7]	pos=	230	score=0.009150	unnormalised=1.1722000000	GPPLSAAAA
8]	pos=	-1	score=0.818414	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_007

AKFVAAWTLKAAAGGSVTVTAFAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEVDIPAVAAAAAIFLIGFALFAAAAAVEIPEFFPYAAAAAKLPHWPTLAAAAAPPMVEGAAAEGDDG
DAAAAADPPFNSILFALLAAAAAANFKSLSSTEFIPNLFAAAAASDREVRGSALYEASTAAAAATSSFVTNTTVGIEL
PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFERADSFAAAAAVFSDEIPASQDMPTNTAAAAAGEGNWTS
DGGTFK

Epitopes of protein 6 -----

0]	pos=	95	score=0.037638	unnormalised=5.0192000000	IPAVAAAAA
1]	pos=	120	score=0.006151	unnormalised=0.8202000000	IPEFFPYAA
2]	pos=	123	score=0.036836	unnormalised=4.9122000000	FPFYAAAAA
3]	pos=	133	score=0.008183	unnormalised=1.0912000000	LPHWPTLA
4]	pos=	137	score=0.026660	unnormalised=3.5552000000	TPTLAAAAA
5]	pos=	140	score=0.009983	unnormalised=1.3312000000	LAAAAAPPM
6]	pos=	145	score=0.000774	unnormalised=0.1032000000	APPMVEGAA
7]	pos=	196	score=0.018749	unnormalised=2.5002000000	IPNLFAAAA
8]	pos=	221	score=0.013529	unnormalised=1.8042000000	AAAAATSSF
9]	pos=	296	score=0.015269	unnormalised=2.0362000000	IPASQDMPT
10]	pos=	302	score=0.040038	unnormalised=5.3392000000	MPTNTAAAA
11]	pos=	-1	score=0.786190	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_008

AKFVAAWTLKAAAGGSYTIPTWTELMAAAAVTVTAFAWAAAAAESLWVG VYRAAAAIFLIGFALFAAAAAANVFDLEGIF
RAAAAAAKLPHWPTLAAAAAVEIPEFFPYAAAAAGTDNEIFLTAAAAAALAEVDIPAVAAAAASDREVRGSALYEAST
AAAAATSSFVTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKSLSSTEFIPNLFAAAAAPPMVEGAAAEGDDG
AAAAAVFSDEIPASQDMPTNTAAAAADGKNKETFERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
DGGTFK

Epitopes of protein 7 -----

0]	pos=	87	score=0.008238	unnormalised=1.0912000000	LPHWPTLA
1]	pos=	91	score=0.026840	unnormalised=3.5552000000	TPTLAAAAA
2]	pos=	102	score=0.006192	unnormalised=0.8202000000	IPEFFPYAA
3]	pos=	105	score=0.037084	unnormalised=4.9122000000	FPFYAAAAA
4]	pos=	136	score=0.037892	unnormalised=5.0192000000	IPAVAAAAA
5]	pos=	160	score=0.013621	unnormalised=1.8042000000	AAAAATSSF
6]	pos=	215	score=0.018875	unnormalised=2.5002000000	IPNLFAAAA

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_009

Epitopes of protein 8 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_010

Epitopes of protein 9 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_011

Epitopes of protein 10 -----

25

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_012

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGAGAGGAGAGGAGAGGAGAGAGGAGAGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAANKPPM

Epitopes of protein 11 -----

0]	pos=	13	score=0.005862	unnormalised=0.6622000000	LPQQLTEA
1]	pos=	73	score=0.002667	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	92	score=0.017106	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.013875	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	247	score=0.021019	unnormalised=2.3742000000	APTGSWFSA
5]	pos=	255	score=0.002649	unnormalised=0.2992000000	APQPAPENA
6]	pos=	291	score=0.008660	unnormalised=0.9782000000	APQPGDNST
7]	pos=	-1	score=0.928163	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_013

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAANKPPM

Epitopes of protein 12 -----

0]	pos=	13	score=0.005862	unnormalised=0.6622000000	LPQQLTEA
1]	pos=	73	score=0.002667	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	92	score=0.017106	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.013875	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	247	score=0.021019	unnormalised=2.3742000000	APTGSWFSA
5]	pos=	255	score=0.002649	unnormalised=0.2992000000	APQPAPENA
6]	pos=	291	score=0.008660	unnormalised=0.9782000000	APQPGDNST
7]	pos=	-1	score=0.928163	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_014

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGNTTVGIELPDAFKCEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRSALYEASTGPGPGELEI

Epitopes of protein 13 -----

0]	pos=	13	score=0.006040	unnormalised=0.6622000000	LPQQLTEA
1]	pos=	73	score=0.002747	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	92	score=0.017623	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.014294	unnormalised=1.5672000000	GPGLPQQQL
4]	pos=	242	score=0.003057	unnormalised=0.3352000000	GPSPGNTTV
5]	pos=	-1	score=0.956238	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_015

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMEFVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGNCKSQGAVAGEPRQIPKIQNFTRTQKSCIEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRSAL

0]	pos=	13	score=0.006058	unnormalised=0.6622000000	LPQGQLTEA
1]	pos=	73	score=0.002756	unnormalised=0.3012000000	RPGAPGSG
2]	pos=	92	score=0.017677	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	216	score=0.014338	unnormalised=1.5672000000	GPGLPQQGL
4]	pos=	-1	score=0.959171	unnormalised=104.8410000000	non-binding event

AKFVAAWTLKAAAGGSANQGQLADIGVPPAPVAAPARRTRKPPQSPESLEECDSELEIKRYKNRVASEAAAKGPGTGP
NGLGEKGDTSPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGSGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGYTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

0]	pos=	27	score=0.033098	unnormalised=3.8422000000	VPQPAPVAA
1]	pos=	29	score=0.031022	unnormalised=3.6012000000	QPAPVAAPA
2]	pos=	121	score=0.002595	unnormalised=0.3012000000	RPGAPGGSG
3]	pos=	140	score=0.016645	unnormalised=1.9322000000	RPQKRPSCI
4]	pos=	264	score=0.013500	unnormalised=1.5672000000	GPGLPQQQL
5]	pos=	-1	score=0.903140	unnormalised=104.8410000000	non-binding event

Read class II peptide list from file? NO

Allele: DRB1_0101
Pseudo sequence: KAF AHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

APPHALSGGSDQATRVYQDLGGPSQAPEAAAKPGTGPNGLGEGKDTSGPEGSGSGSPQRRGDNHGRGRGRGRGRGGG
RPGAPGGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGA
GGGAGGAGGAGAGGGAGAEAAAKRVLVVVLLGPGPGTMEVSVAFGPGPGACMLSAPELKGPGPGPLPQQQLTAYGPGPG
FLDKGTYTLGPGPGVFSDEIPASQDMPTNTEAAAKPPMVEGAAAEAGDDGDPGPGSDREVRGSALEYEAST

0]	pos= 16	score=0.038938	unnormalised=3.2345600000	YQDLGGPSQ
1]	pos= 188	score=0.020147	unnormalised=1.6735600000	VVLLGPGPG
2]	pos= 189	score=0.010251	unnormalised=0.8515600000	VLLGPGPGT
3]	pos= 205	score=0.035266	unnormalised=2.9295600000	FGPGPGACM
4]	pos= 218	score=0.010733	unnormalised=0.8915600000	LEKGPGPGL
5]	pos= 246	score=0.088126	unnormalised=7.3205600000	YTLGPGPGV
6]	pos= -1	score=0.796539	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_002

AKFVAAWTLKAAAGGSCPSLDVDSIIPRTDPVLHEDEAAAKPGTGPNGLGEKGDTSGPESGGSGPQRRGGDNHGRGR
GRGRGRGGGRPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGAGGAGG
AGAGGGAGAGGGAGGAGAGAGGGAGAGAAAKRVLVSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGG
LTAYGPGPGFLDKGTYTLGPGPGPLEKQLFYIIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVGRS

Epitopes of protein 1 -----

0]	pos=	2	score=0.041679	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.005684	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	197	score=0.020609	unnormalised=1.6735600000	VVLLGPGPG
3]	pos=	198	score=0.010487	unnormalised=0.8515600000	VLLGPGPGT
4]	pos=	214	score=0.036076	unnormalised=2.9295600000	FGPGPGACM
5]	pos=	227	score=0.010979	unnormalised=0.8915600000	LEKGPGL
6]	pos=	269	score=0.034894	unnormalised=2.8335600000	FYYIGTMLP
7]	pos=	272	score=0.024759	unnormalised=2.0105600000	IGTMLPNTR
8]	pos=	-1	score=0.814832	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_003

AKFVAAWTLKAAAGGSDLERGPPGPRRPPRPPPLSGPGTGPNGLGEKGDTSGPESGGSGPQRRGGDNHGRGRGRGRGR
GGGRPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGAGGAGAGGAGG
AGAGGGAGGAGGAGAGGGAGAGAAAKVPKPNVEVWGPGRQIPKIQNFPGPGMVFLQTHIFAEVLKDGPGPGSALIAG
GSILQTNFKEAAKNEIFLTKKMTVCQA

Epitopes of protein 2 -----

0]	pos=	2	score=0.047318	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.006453	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	194	score=0.021160	unnormalised=1.5135600000	WGPGRQI
3]	pos=	-1	score=0.925068	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_004

AKFVAAWTLKAAAGSGPGTGPNGLGEKGDTSGPESGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSPRHRD
GVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGAGAGGAGAGGGAGAGGAGAGGAGAGGGA
GAEEAAKFLDKGTYTLGPGPGTVMETIAGLYGPGPGTSKFLMGTYKRVTEKGPGRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 3 -----

0]	pos=	2	score=0.040812	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.005566	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	8	score=0.040800	unnormalised=3.3835600000	LKAAAGGSG
3]	pos=	173	score=0.077469	unnormalised=6.4245600000	YTLGPGPGT
4]	pos=	198	score=0.011366	unnormalised=0.9425600000	FLMGTYKRV
5]	pos=	218	score=0.026113	unnormalised=2.1655600000	LLVSGAEAA
6]	pos=	-1	score=0.797874	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_005

AKFVAAWTLKAAAGGSDQATRVYQDLGGPSQAPPIDELAQQHASGEGPGINPISKTA AAAALERGPPGPRRPPRPPPLS
AAAAAGTYKRVTEKAAAAADPYQVPVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSSTEFIPNLFAAAAAASCTASVSTVVSATAAAAAATPDIL
PVVTRNLNAI

Epitopes of protein 4 -----

0]	pos=	2	score=0.012370	unnormalised=3.3845600000	FVAAWTLKA
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1]	pos=	6	score=0.001687	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	22	score=0.011821	unnormalised=3.2345600000	YQDLGGPSQ
3]	pos=	53	score=0.019712	unnormalised=5.3935600000	ISKTA AAAA
4]	pos=	78	score=0.025724	unnormalised=7.0385600000	LSAAAAAGT
5]	pos=	87	score=0.007651	unnormalised=2.0935600000	YKRVTEKAA
6]	pos=	105	score=0.029400	unnormalised=8.0445600000	FVAAAAAYQ
7]	pos=	106	score=0.023465	unnormalised=6.4205600000	VAAAAAYQA
8]	pos=	112	score=0.030968	unnormalised=8.4735600000	YQAYAAPQL
9]	pos=	120	score=0.031103	unnormalised=8.5105600000	LFAAAAAAT
10]	pos=	121	score=0.045298	unnormalised=12.3945600000	FAAAAAATV
11]	pos=	129	score=0.036998	unnormalised=10.1235600000	VKGMQSYAA
12]	pos=	132	score=0.017347	unnormalised=4.7465600000	MQSYAAAAA
13]	pos=	135	score=0.044293	unnormalised=12.1195600000	YAAAAAIAL
14]	pos=	143	score=0.015915	unnormalised=4.3545600000	LYLQQNWAA
15]	pos=	145	score=0.022193	unnormalised=6.0725600000	LQQNWAAAA
16]	pos=	149	score=0.042831	unnormalised=11.7195600000	WAAAAANST
17]	pos=	162	score=0.002056	unnormalised=0.5625600000	VVRAQGLDA
18]	pos=	178	score=0.015688	unnormalised=4.2925600000	FNSILFALL
19]	pos=	182	score=0.037203	unnormalised=10.1795600000	LFALLAAAA
20]	pos=	183	score=0.054936	unnormalised=15.0315600000	FALLAAAAA
21]	pos=	185	score=0.034963	unnormalised=9.5665600000	LLAAAAAAA
22]	pos=	186	score=0.032912	unnormalised=9.0055600000	LAAAAAAA
23]	pos=	196	score=0.028626	unnormalised=7.8325600000	FKSLSSSTEF
24]	pos=	205	score=0.000784	unnormalised=0.2145600000	IPNLFAAAA
25]	pos=	208	score=0.030533	unnormalised=8.3545600000	LFAAAAAAS
26]	pos=	209	score=0.024437	unnormalised=6.6865600000	FAAAAAASC
27]	pos=	218	score=0.027924	unnormalised=7.6405600000	FTASVSTTV
28]	pos=	222	score=0.006665	unnormalised=1.8235600000	VSTVVSATA
29]	pos=	225	score=0.020709	unnormalised=5.6665600000	VVSATAAAA
30]	pos=	226	score=0.021963	unnormalised=6.0095600000	VSATAAAA
31]	pos=	-1	score=0.241823	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_006

AKFVAAWTLKAAAGGSPGIDELAQKHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQQNWAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRPPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 5 -----

0]	pos=	2	score=0.013245	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.001806	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	36	score=0.021107	unnormalised=5.3935600000	ISKTA AAAA
3]	pos=	51	score=0.012658	unnormalised=3.2345600000	YQDLGGPSQ
4]	pos=	67	score=0.033159	unnormalised=8.4735600000	YQAYAAPQL
5]	pos=	75	score=0.023728	unnormalised=6.0635600000	LFAAAAAGT
6]	pos=	76	score=0.008823	unnormalised=2.2545600000	FAAAAAAGTY
7]	pos=	84	score=0.008193	unnormalised=2.0935600000	YKRVTEKAA
8]	pos=	99	score=0.016798	unnormalised=4.2925600000	FNSILFALL
9]	pos=	103	score=0.039836	unnormalised=10.1795600000	LFALLAAAA
10]	pos=	104	score=0.058823	unnormalised=15.0315600000	FALLAAAAA
11]	pos=	106	score=0.037437	unnormalised=9.5665600000	LLAAAAAAA
12]	pos=	107	score=0.035241	unnormalised=9.0055600000	LAAAAAAA
13]	pos=	117	score=0.030651	unnormalised=7.8325600000	FKSLSSSTEF
14]	pos=	126	score=0.000840	unnormalised=0.2145600000	IPNLFAAAA
15]	pos=	129	score=0.021917	unnormalised=5.6005600000	LFAAAAANS
16]	pos=	130	score=0.052233	unnormalised=13.3475600000	FAAAAAANST
17]	pos=	143	score=0.002201	unnormalised=0.5625600000	VVRAQGLDA
18]	pos=	163	score=0.007860	unnormalised=2.0085600000	VTRNLNAIA
19]	pos=	167	score=0.044727	unnormalised=11.4295600000	LNAIAAAAA
20]	pos=	170	score=0.030585	unnormalised=7.8155600000	IAAAAAIAL
21]	pos=	178	score=0.017041	unnormalised=4.3545600000	LYLQQNWAA

22]	pos= 180	score=0.023764	unnormalised=6.0725600000	LQQNWAAAA
23]	pos= 184	score=0.042133	unnormalised=10.7665600000	WAAAAAATV
24]	pos= 192	score=0.039616	unnormalised=10.1235600000	VKGMQSYAA
25]	pos= 195	score=0.018575	unnormalised=4.7465600000	MQSYAAAAA
26]	pos= 198	score=0.007269	unnormalised=1.8575600000	YAAAAADPY
27]	pos= 210	score=0.024394	unnormalised=6.2335600000	FVAAAAALE
28]	pos= 233	score=0.036509	unnormalised=9.3295600000	LSAAAAAAS
29]	pos= 243	score=0.029900	unnormalised=7.6405600000	FTASVSTVV
30]	pos= -1	score=0.258934	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_007

AKFVAAWTLKAAAGGSVTVTAFWAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEVPDIPAVAAAAAIFLIGFALFAAAAAVEIPEFPFYAAAAAKLPHWPTLAAAAAPPMVEGAAAGDDG
DAAAAADPPFNLSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAASDREVRGSALYEASTAAAAATSSSFVTNTTVGIEL
PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFHERADSFAAAAAVFSDEIPASQDMPNTAAAAAGEGNWTSLSL
DGGTFFK

Epitopes of protein 6 -----

0]	pos= 2	score=0.010488	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001430	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 8	score=0.021189	unnormalised=6.8375600000	LKAAAGGSV
3]	pos= 21	score=0.013745	unnormalised=4.4355600000	FWAWAAAAA
4]	pos= 22	score=0.016894	unnormalised=5.4515600000	WAWAAAAAY
5]	pos= 24	score=0.032190	unnormalised=10.3875600000	WAAAAAYTI
6]	pos= 34	score=0.024808	unnormalised=8.0055600000	WTELMAAAA
7]	pos= 35	score=0.002744	unnormalised=0.8855600000	TELMAAAAA
8]	pos= 37	score=0.025763	unnormalised=8.3135600000	LMAAAAANV
9]	pos= 38	score=0.003652	unnormalised=1.1785600000	MAAAAAANV
10]	pos= 51	score=0.018375	unnormalised=5.9295600000	IFRAAAAAAG
11]	pos= 52	score=0.039082	unnormalised=12.6115600000	FRAAAAAAGT
12]	pos= 64	score=0.012608	unnormalised=4.0685600000	IFLTAAAAA
13]	pos= 65	score=0.023002	unnormalised=7.4225600000	FLTAAAAAA
14]	pos= 80	score=0.024564	unnormalised=7.9265600000	VYRAAAAAA
15]	pos= 81	score=0.047189	unnormalised=15.2275600000	YRAAAAAAL
16]	pos= 95	score=0.006698	unnormalised=2.1615600000	IPAVAAAAA
17]	pos= 98	score=0.014458	unnormalised=4.6655600000	VAAAAAIFL
18]	pos= 105	score=0.008564	unnormalised=2.7635600000	FLIGFALFA
19]	pos= 109	score=0.026751	unnormalised=8.6325600000	FALFAAAAA
20]	pos= 112	score=0.029212	unnormalised=9.4265600000	FAAAAAVEI
21]	pos= 125	score=0.039094	unnormalised=12.6155600000	FYAAAAAKL
22]	pos= 126	score=0.012593	unnormalised=4.0635600000	YAAAAAKLP
23]	pos= 136	score=0.022388	unnormalised=7.2245600000	WTPTLAAAA
24]	pos= 140	score=0.027870	unnormalised=8.9935600000	LAAAAAPPM
25]	pos= 169	score=0.013302	unnormalised=4.2925600000	FNSILFALL
26]	pos= 173	score=0.031545	unnormalised=10.1795600000	LFALLAAAA
27]	pos= 174	score=0.046581	unnormalised=15.0315600000	FALLAAAAA
28]	pos= 176	score=0.029646	unnormalised=9.5665600000	LLAAAAAAA
29]	pos= 177	score=0.027907	unnormalised=9.0055600000	LAAAAAAA
30]	pos= 187	score=0.024272	unnormalised=7.8325600000	FKSLSSSTEF
31]	pos= 196	score=0.000665	unnormalised=0.2145600000	IPNLFAAAA
32]	pos= 200	score=0.018016	unnormalised=5.8135600000	FAAAAAASDR
33]	pos= 215	score=0.003708	unnormalised=1.1965600000	LYEASTAAA
34]	pos= 216	score=0.029215	unnormalised=9.4275600000	YEASTAAAA
35]	pos= 229	score=0.012552	unnormalised=4.0505600000	FVTNTTVGI
36]	pos= 237	score=0.011229	unnormalised=3.6235600000	IELPAAAAA
37]	pos= 261	score=0.005338	unnormalised=1.7225600000	VLHEDAAAA
38]	pos= 262	score=0.000336	unnormalised=0.1085600000	LHEDAAAAA
39]	pos= 285	score=0.035289	unnormalised=11.3875600000	FAAAAAVFS
40]	pos= -1	score=0.205047	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_010

Epitopes of protein 9 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_011

Epitopes of protein 10 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_012

Epitopes of protein 11 -----

32

4]	pos= 211	score=0.009830	unnormalised=0.8915600000	LEKGP GPGL
5]	pos= 239	score=0.069191	unnormalised=6.2755600000	YTLGP GPGA
6]	pos= 241	score=0.002608	unnormalised=0.2365600000	LGPG GPAPT
7]	pos= 264	score=0.093425	unnormalised=8.4735600000	YQAYAAPQL
8]	pos= -1	score=0.729530	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_013

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGP GPGLPQQQLTAYGPGPGFLDKGTY
TLGPGGPAPTGSWFSAPQPAPENAYQAYAAPQLFPVSDITQNNQNTNAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 12 -----

0]	pos= 5	score=0.035276	unnormalised=3.1995600000	LSGGSPEPL
1]	pos= 181	score=0.018452	unnormalised=1.6735600000	VVLLGPGPG
2]	pos= 182	score=0.009389	unnormalised=0.8515600000	VLLGPGPGT
3]	pos= 198	score=0.032300	unnormalised=2.9295600000	FGPGPGACM
4]	pos= 211	score=0.009830	unnormalised=0.8915600000	LEKGP GPGL
5]	pos= 239	score=0.069191	unnormalised=6.2755600000	YTLGP GPGA
6]	pos= 241	score=0.002608	unnormalised=0.2365600000	LGPG GPAPT
7]	pos= 264	score=0.093425	unnormalised=8.4735600000	YQAYAAPQL
8]	pos= -1	score=0.729530	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_014

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGP GPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGNTTVGIELPDAFKCEAAAKPPMVEGAAEGDDGDGPGPGSDREVRGSALYEASTGPGPGELEI

Epitopes of protein 13 -----

0]	pos= 5	score=0.040762	unnormalised=3.1995600000	LSGGSPEPL
1]	pos= 181	score=0.021321	unnormalised=1.6735600000	VVLLGPGPG
2]	pos= 182	score=0.010849	unnormalised=0.8515600000	VLLGPGPGT
3]	pos= 198	score=0.037323	unnormalised=2.9295600000	FGPGPGACM
4]	pos= 211	score=0.011358	unnormalised=0.8915600000	LEKGP GPGL
5]	pos= 239	score=0.033437	unnormalised=2.6245600000	YTLGP GPGN
6]	pos= 258	score=0.001969	unnormalised=0.1545600000	FKCEAAAKP
7]	pos= -1	score=0.842980	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_015

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGP GPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGNKCSQGA VAGEPRQIPKIQNFTRTQKSCIEAAAKPPMVEGAAEGDDGDGPGPGSDREVRGSA

Epitopes of protein 14 -----

0]	pos= 5	score=0.040843	unnormalised=3.1995600000	LSGGSPEPL
1]	pos= 181	score=0.021363	unnormalised=1.6735600000	VVLLGPGPG
2]	pos= 182	score=0.010870	unnormalised=0.8515600000	VLLGPGPGT
3]	pos= 198	score=0.037396	unnormalised=2.9295600000	FGPGPGACM
4]	pos= 211	score=0.011381	unnormalised=0.8915600000	LEKGP GPGL
5]	pos= 239	score=0.033503	unnormalised=2.6245600000	YTLGP GPGN
6]	pos= -1	score=0.844644	unnormalised=66.1680000000	non-binding event

4]	pos= 214	score=0.036076	unnormalised=2.9295600000	FGPGPGACM
5]	pos= 227	score=0.010979	unnormalised=0.8915600000	LEKPGPGPL
6]	pos= 269	score=0.034894	unnormalised=2.8335600000	FYYIGTMLP
7]	pos= 272	score=0.024759	unnormalised=2.0105600000	IGTMLPNTR
8]	pos= -1	score=0.814832	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_003

AKFVAAWTLKAAAGGSDLERGPPGPRRPPRGPPLSGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGR
GGGRPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGG
AGAGGGAGGAGGAGGAGGAGAAAKVPKNVEVWGPGRQIPKIQNFGPGPMVFLQTHIFAEVLKDGPGPSALIA
GSILQTNFKEAAKNEIFLTKKMTVCQA

Epitopes of protein 2 -----

0]	pos= 2	score=0.047318	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.006453	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 194	score=0.021160	unnormalised=1.5135600000	WGPGRQI
3]	pos= -1	score=0.925068	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_004

AKFVAAWTLKAAAGGSGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGGA
GAEEAAKFLDKGTYTLGPGPGTVMIEIAGLYGPGPTSKFLMGTYKRVTEKGPGRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 3 -----

0]	pos= 2	score=0.040812	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.005566	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 8	score=0.040800	unnormalised=3.3835600000	LKAAAGGSG
3]	pos= 173	score=0.077469	unnormalised=6.4245600000	YTLGPGPGT
4]	pos= 198	score=0.011366	unnormalised=0.9425600000	FLMGTYKRV
5]	pos= 218	score=0.026113	unnormalised=2.1655600000	LLVSGAEAA
6]	pos= -1	score=0.797874	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_005

AKFVAAWTLKAAAGGSDQATRVYQDLGGPSQAPPIDELAQHASGEGPGINPISKTAATAAALERGPPGPRRPPRGPPLS
AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAASCFTASVSTTVVSATAAAAAATPDIL
PVVTRNLNAI

Epitopes of protein 4 -----

0]	pos= 2	score=0.012370	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001687	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 22	score=0.011821	unnormalised=3.2345600000	YQDLGGPSQ
3]	pos= 53	score=0.019712	unnormalised=5.3935600000	ISKTAATAA
4]	pos= 78	score=0.025724	unnormalised=7.0385600000	LSAAAAAGT
5]	pos= 87	score=0.007651	unnormalised=2.0935600000	YKRVTEKAA
6]	pos= 105	score=0.029400	unnormalised=8.0445600000	FVAAAAAYQ
7]	pos= 106	score=0.023465	unnormalised=6.4205600000	VAAAAAYQA
8]	pos= 112	score=0.030968	unnormalised=8.4735600000	YQAYAAPQL
9]	pos= 120	score=0.031103	unnormalised=8.5105600000	LFAAAAAAT
10]	pos= 121	score=0.045298	unnormalised=12.3945600000	FAAAAAATV
11]	pos= 129	score=0.036998	unnormalised=10.1235600000	VKGMQSYAA
12]	pos= 132	score=0.017347	unnormalised=4.7465600000	MQSYAAAAA
13]	pos= 135	score=0.044293	unnormalised=12.1195600000	YAAAAAIAL

14]	pos= 143	score=0.015915	unnormalised=4.3545600000	LYLQQNWAA
15]	pos= 145	score=0.022193	unnormalised=6.0725600000	LQQNWAAAA
16]	pos= 149	score=0.042831	unnormalised=11.7195600000	WAAAAANST
17]	pos= 162	score=0.002056	unnormalised=0.5625600000	VVRAQGLDA
18]	pos= 178	score=0.015688	unnormalised=4.2925600000	FNSILFALL
19]	pos= 182	score=0.037203	unnormalised=10.1795600000	LFALLAAAA
20]	pos= 183	score=0.054936	unnormalised=15.0315600000	FALLAAAAA
21]	pos= 185	score=0.034963	unnormalised=9.5665600000	LLAAAAAAA
22]	pos= 186	score=0.032912	unnormalised=9.0055600000	LAAAAAAA
23]	pos= 196	score=0.028626	unnormalised=7.8325600000	FKSLSTEF
24]	pos= 205	score=0.000784	unnormalised=0.2145600000	IPNLFAAAA
25]	pos= 208	score=0.030533	unnormalised=8.3545600000	LFAAAAAAS
26]	pos= 209	score=0.024437	unnormalised=6.6865600000	FAAAAAASC
27]	pos= 218	score=0.027924	unnormalised=7.6405600000	FTASVSTVV
28]	pos= 222	score=0.006665	unnormalised=1.8235600000	VSTVVSATA
29]	pos= 225	score=0.020709	unnormalised=5.6665600000	VVSATAAAA
30]	pos= 226	score=0.021963	unnormalised=6.0095600000	VSATAAAAA
31]	pos= -1	score=0.241823	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_006

AKFVAAWTLKAAAGGSPGIDELAQGHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQKNWAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRPPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 5 -----

0]	pos= 2	score=0.013245	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001806	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 36	score=0.021107	unnormalised=5.3935600000	ISKTA AAAA
3]	pos= 51	score=0.012658	unnormalised=3.2345600000	YQDLGGPSQ
4]	pos= 67	score=0.033159	unnormalised=8.4735600000	YQAYAAPQL
5]	pos= 75	score=0.023728	unnormalised=6.0635600000	LFAAAAAAGT
6]	pos= 76	score=0.008823	unnormalised=2.2545600000	FAAAAAAGTY
7]	pos= 84	score=0.008193	unnormalised=2.0935600000	YKRVTEKAA
8]	pos= 99	score=0.016798	unnormalised=4.2925600000	FNSILFALL
9]	pos= 103	score=0.039836	unnormalised=10.1795600000	LFALLAAAA
10]	pos= 104	score=0.058823	unnormalised=15.0315600000	FALLAAAAA
11]	pos= 106	score=0.037437	unnormalised=9.5665600000	LLAAAAAAA
12]	pos= 107	score=0.035241	unnormalised=9.0055600000	LAAAAAAA
13]	pos= 117	score=0.030651	unnormalised=7.8325600000	FKSLSTEF
14]	pos= 126	score=0.000840	unnormalised=0.2145600000	IPNLFAAAA
15]	pos= 129	score=0.021917	unnormalised=5.6005600000	LFAAAAAANS
16]	pos= 130	score=0.052233	unnormalised=13.3475600000	FAAAAAANST
17]	pos= 143	score=0.002201	unnormalised=0.5625600000	VVRAQGLDA
18]	pos= 163	score=0.007860	unnormalised=2.0085600000	VTRNLNAIA
19]	pos= 167	score=0.044727	unnormalised=11.4295600000	LNAIAAAAA
20]	pos= 170	score=0.030585	unnormalised=7.8155600000	IAAAAAIAL
21]	pos= 178	score=0.017041	unnormalised=4.3545600000	LYLQQNWAA
22]	pos= 180	score=0.023764	unnormalised=6.0725600000	LQQNWAAAA
23]	pos= 184	score=0.042133	unnormalised=10.7665600000	WAAAAAATV
24]	pos= 192	score=0.039616	unnormalised=10.1235600000	VKGMQSYAA
25]	pos= 195	score=0.018575	unnormalised=4.7465600000	MQSYAAAAA
26]	pos= 198	score=0.007269	unnormalised=1.8575600000	YAAAAADPY
27]	pos= 210	score=0.024394	unnormalised=6.2335600000	FVAAAAALE
28]	pos= 233	score=0.036509	unnormalised=9.3295600000	LSAAAAAAS
29]	pos= 243	score=0.029900	unnormalised=7.6405600000	FTASVSTVV
30]	pos= -1	score=0.258934	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_007

AKFVAAWTLKAAAGGSVTVTAFAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
 VYRAAAAAALAEPVDIPAVAAAAAIFLIGFALFAAAAAVEIPEFFFYAAAAAKLPHWPTLAAAAAPPMEGAAAAEGDDG
 DAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAASDREVRGSALYEASTAAAAATSSFVTNTTVGIEL
 PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFERADSFAAAAAVFSDEIPASQDMPNTNTAAAAAGEGNWTS
 L
 DGGTFK

Epitopes of protein 6 -----

0]	pos=	2	score=0.010488	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.001430	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	8	score=0.021189	unnormalised=6.8375600000	LKAAAGGSV
3]	pos=	21	score=0.013745	unnormalised=4.4355600000	FWAAAAAAA
4]	pos=	22	score=0.016894	unnormalised=5.4515600000	WAAAAAAAY
5]	pos=	24	score=0.032190	unnormalised=10.3875600000	WAAAAAYTI
6]	pos=	34	score=0.024808	unnormalised=8.0055600000	WTELMAAAA
7]	pos=	35	score=0.002744	unnormalised=0.8855600000	TELMAAAAA
8]	pos=	37	score=0.025763	unnormalised=8.3135600000	LMAAAAAANV
9]	pos=	38	score=0.003652	unnormalised=1.1785600000	MAAAAAANVF
10]	pos=	51	score=0.018375	unnormalised=5.9295600000	IFRAAAAAAG
11]	pos=	52	score=0.039082	unnormalised=12.6115600000	FRAAAAAAGT
12]	pos=	64	score=0.012608	unnormalised=4.0685600000	IFLTAAAAA
13]	pos=	65	score=0.023002	unnormalised=7.4225600000	FLTAAAAAA
14]	pos=	80	score=0.024564	unnormalised=7.9265600000	VYRAAAAAA
15]	pos=	81	score=0.047189	unnormalised=15.2275600000	YRAAAAAAL
16]	pos=	95	score=0.006698	unnormalised=2.1615600000	IPAVAAAAA
17]	pos=	98	score=0.014458	unnormalised=4.6655600000	VAAAAAIFL
18]	pos=	105	score=0.008564	unnormalised=2.7635600000	FLIGFALFA
19]	pos=	109	score=0.026751	unnormalised=8.6325600000	FALFAAAAA
20]	pos=	112	score=0.029212	unnormalised=9.4265600000	FAAAAAVEI
21]	pos=	125	score=0.039094	unnormalised=12.6155600000	FYAAAAAKL
22]	pos=	126	score=0.012593	unnormalised=4.0635600000	YAAAAAKLP
23]	pos=	136	score=0.022388	unnormalised=7.2245600000	WPTTLAAAA
24]	pos=	140	score=0.027870	unnormalised=8.9935600000	LAAAAAPP
25]	pos=	169	score=0.013302	unnormalised=4.2925600000	FNSILFALL
26]	pos=	173	score=0.031545	unnormalised=10.1795600000	LFALLAAAA
27]	pos=	174	score=0.046581	unnormalised=15.0315600000	FALLAAAAA
28]	pos=	176	score=0.029646	unnormalised=9.5665600000	LLAAAAAAA
29]	pos=	177	score=0.027907	unnormalised=9.0055600000	LAAAAAAA
30]	pos=	187	score=0.024272	unnormalised=7.8325600000	FKSLSTEF
31]	pos=	196	score=0.000665	unnormalised=0.2145600000	IPNLFAAAA
32]	pos=	200	score=0.018016	unnormalised=5.8135600000	FAAAAASDR
33]	pos=	215	score=0.003708	unnormalised=1.1965600000	LYEASTAAA
34]	pos=	216	score=0.029215	unnormalised=9.4275600000	YEASTAAA
35]	pos=	229	score=0.012552	unnormalised=4.0505600000	FVTNTTVGI
36]	pos=	237	score=0.011229	unnormalised=3.6235600000	IELPAAAAA
37]	pos=	261	score=0.005338	unnormalised=1.7225600000	VLHEDAAAA
38]	pos=	262	score=0.000336	unnormalised=0.1085600000	LHEDAAAAA
39]	pos=	285	score=0.035289	unnormalised=11.3875600000	FAAAAVFS
40]	pos=	-1	score=0.205047	unnormalised=66.1680000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_008

AKFVAAWTLKAAAGGSYTIPWTELMAAAAAVTVTAFAWAAAAAESLWVG VYRAAAAAIFLIGFALFAAAAAANVFDLEGIF
 RAAAAAKLPHWPTLAAAAAVEIPEFFFYAAAAAGTDNEIFLTAAAAAALAEPVDIPAVAAAAASDREVRGSALYEAST
 AAAAATSSFVTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAPPMEGAAAAEGDDGD
 AAAA AVFSDEIPASQDMPNTNTAAAAADGKNKETFERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
 L
 DGGTFK

Epitopes of protein 7 -----

0]	pos=	2	score=0.011091	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.001513	unnormalised=0.4615600000	WTLKAAAGG

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_009

Epitopes of protein 8 -----

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-005639_5_QJs2n4D23IFC.FSA_1_010

Epitopes of protein 9 -----

Epitopes of protein 10 -----

Epitopes of protein 11 -----

1]	pos=	6	score=0.005220	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	8	score=0.065517	unnormalised=5.7925600000	LKAAAGGSA
3]	pos=	61	score=0.018793	unnormalised=1.6615600000	YKNRVASEA
4]	pos=	65	score=0.009609	unnormalised=0.8495600000	VASEAAAKG
5]	pos=	229	score=0.018929	unnormalised=1.6735600000	VVLLGPGPG
6]	pos=	230	score=0.009632	unnormalised=0.8515600000	VLLGPGPGT
7]	pos=	246	score=0.033135	unnormalised=2.9295600000	FGPGPGACM
8]	pos=	259	score=0.010084	unnormalised=0.8915600000	LEKGPGPGL
9]	pos=	300	score=0.042409	unnormalised=3.7495600000	VTTQGPEAA
10]	pos=	-1	score=0.748392	unnormalised=66.1680000000	non-binding event
